

YAML

YAML (YAML Ain't Markup Language) is a human-readable data serialization format that is often used for configuration files, data exchange between programming languages, and defining data structures in a readable way. It is designed to be simple and intuitive, making it easier for humans to read and write compared to other formats like JSON or XML.

Key Features of YAML:

1. **Human-Readable:** YAML uses indentation and a minimalist syntax to make files easy to read and write.
2. **Data Serialization:** It can represent data structures such as lists, dictionaries, and scalars (simple values like strings or numbers).
3. **Language-Independent:** YAML is not tied to any specific programming language, and many languages provide libraries to parse and generate YAML.
4. **Flexible and Expressive:** Supports complex data types, references (anchors and aliases), and extensible tags for custom types.
5. **Widely Used:** Commonly found in configuration files for tools like Docker Compose, Kubernetes, Ansible, GitLab CI, and more.

YAML Data Types

1. Scalars (Basic Types)

- **String:** Text values; used for names, messages, descriptions.
- **Number:** Integers or floats; used for IDs, quantities, measurements.
- **Boolean:** Logical values (`true/false`); used for binary states.
- **Null:** Represents absence of value; used for optional fields.

2. Sequences (Lists)

- **Description:** Ordered collections of items.
- **Use:** Lists of related values, e.g., items, names, configurations.

3. Mappings (Dictionaries)

- **Description:** Key-value pairs for hierarchical data.
- **Use:** Store structured information, e.g., settings or attributes.

4. Advanced Features

- **Aliases & Anchors:** Reuse data; avoid duplication in configurations.
- **Multiline Strings:** Handle long text (logs, descriptions).
- **Tags & Types:** Explicitly declare data types (e.g., force number as a string).

scalars

Name: Google

Location: California

CEO: sundar pichai

list

Options:

- search

- workspace

- topics

- voice control

Mappings

Plans:

- paid: business standard

- free: gmail

Multi_line scalar

About google: |

Google is one of the world's largest technology companies, founded on September 4, 1998, by Larry Page and Sergey Brin while they were PhD students at Stanford University. Initially, Google began as a search engine designed to organize and make information accessible and useful. Over the years, it has expanded into various areas of technology, offering a wide range of products and services.

YAML Lint

Paste in your YAML and click "Go" - we'll tell you if it's valid or not, and give you a nice clean UTF-8 version of it.

```
1 ---
2 Name: Google
3 Location: California
4 CEO: sundar pichai
5 Options: -search -workspace -topics -voice control
6 Plans:
7   paid: business standard
8   free: gmail
9 About google: >
10 Google is one of the world's largest technology companies, founded
11 on September 4, 1998, by Larry Page and Sergey Brin while they were PhD
12 students at Stanford University. Initially, Google began as a search engine
13 designed to organize and make information accessible and useful. Over the
14 years, it has expanded into various areas of technology, offering a wide range
15 of products and services.
16
17
18
19
20
```

Go ☒ Reformat (strips comments) ☒ Resolve aliases

Valid YAML!

YAML Syntax

1. Key Syntax Rules

Key-Value Pairs:

Data is written as key: value.

name: sunil

age: 30

Indentation:

Use spaces (not tabs) for nesting data.

Indentation indicates hierarchy.

Comments:

Add comments using #.

This is a comment

key: value

2. Lists (Sequences)

Represented with a dash (-) before each item.

fruits:

- Apple
- Banana
- Cherry

3.Nested Data

Combine lists and key-value pairs for hierarchy

person:

name: John

age: 30

hobbies:

- Reading
- Hiking
- Coding

4.Multiline Strings

Literal Block (|): Keeps line breaks.

description: |

This is line 1.

This is line 2.

description: >

This text

folds into one line.